

Experiment 'p_6_2_b_shift_right_array' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_b_shift_right_array

Natural language statement: Write a method to shift all elements of an array by one to the right and move the last element into the first position. For example, [1, 4, 9, 16, 25] would be transformed into [25, 1, 4, 9, 16].

Method signature: p_6_2_b_shift_right_array(inputs: array<int>) returns (result: array<int>)

Ensures

- `result.Length == inputs.Length`
- `result[0] == inputs[inputs.Length - 1]`
- forall `i :: 1 <= i < inputs.Length ==> result[i] == inputs[i - 1]`

Requires

- `inputs.Length >= 1`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to shift all elements of an array by one to the right and move the last
→ element into the first position. For example, [1, 4, 9, 16, 25] would be
→ transformed into [25, 1, 4, 9, 16].

The signature should be:

p_6_2_b_shift_right_array(inputs: array<int>) returns (result: array<int>)

The method should respect the following contract:

requires inputs.Length >= 1, ensures result.Length == inputs.Length, ensures result[0] ==
→ inputs[inputs.Length - 1], ensures forall i :: 1 <= i < inputs.Length ==> result[
→ i] == inputs[i - 1]

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0DRUTRnUVNyQqx5boD9YLLQAUmA

Model: gpt-5.5-2026-04-23

Created at: 1780571273

Finish reason: stop

Usage: 444 tokens in, and 534 tokens out

```
{
  result := new int[inputs.Length];
  result[0] := inputs[inputs.Length - 1];

  var i := 1;
  while i < inputs.Length
    invariant 1 <= i <= inputs.Length
    invariant result.Length == inputs.Length
    invariant result[0] == inputs[inputs.Length - 1]
    invariant forall k :: 1 <= k < i ==> result[k] == inputs[k - 1]
    decreases inputs.Length - i
  {
    result[i] := inputs[i - 1];
    i := i + 1;
  }
}
```

Final Program

```
method p_6_2_b_shift_right_array(inputs: array<int>) returns (result: array<int>)
  requires inputs.Length >= 1
  ensures result.Length == inputs.Length
  ensures result[0] == inputs[inputs.Length - 1]
  ensures forall i :: 1 <= i < inputs.Length ==> result[i] == inputs[i - 1]
{
  result := new int[inputs.Length];
  result[0] := inputs[inputs.Length - 1];

  var i := 1;
  while i < inputs.Length
    invariant 1 <= i <= inputs.Length
    invariant result.Length == inputs.Length
    invariant result[0] == inputs[inputs.Length - 1]
    invariant forall k :: 1 <= k < i ==> result[k] == inputs[k - 1]
    decreases inputs.Length - i
  {
    result[i] := inputs[i - 1];
    i := i + 1;
  }
}
```

Total Token Usage

Input tokens: 444

Output tokens: 534

Reasoning tokens: 379

Sum of 'total tokens': 978

Experiment Timings

Overall Experiment started at 1780571272977, ended at 1780571302708, lasting 29731ms (29.73 seconds)

Iteration #1 started at 1780571272990, ended at 1780571302708, lasting 29718ms (29.72 seconds)